CPEN 416

Lecture 10: Quantum compilation I

Thursday 15 October 2026

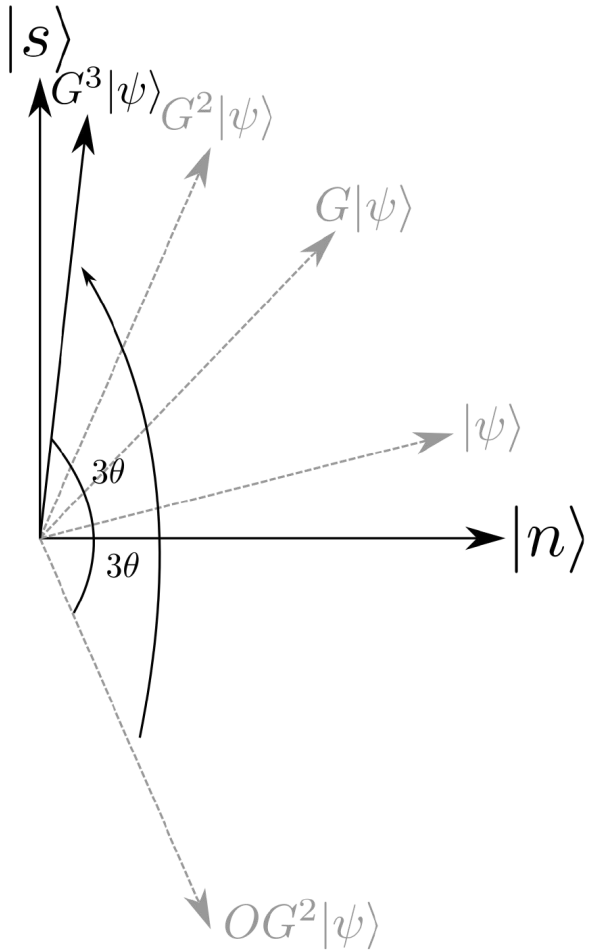
Announcements

- Project details available by tomorrow
- *No formal tutorial Monday*, but Parvin will hold office hours in usual room
- Quiz 5 on Tuesday

Last time

We developed a geometric formulation for Grover's algorithm as a series of reflections implemented by the oracle and diffusion operators.

Grover's algorithm yields a square-root speedup in **query complexity** over classical methods for unstructured search.

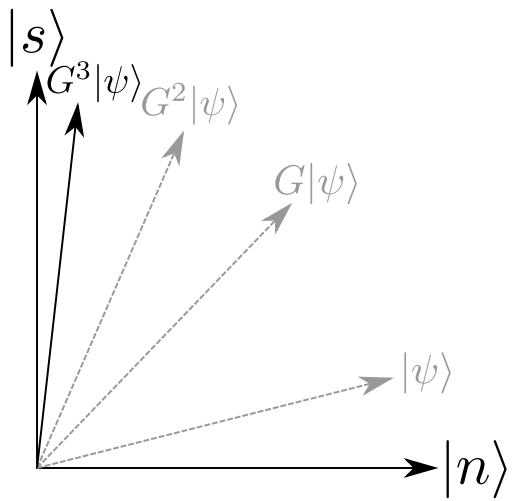


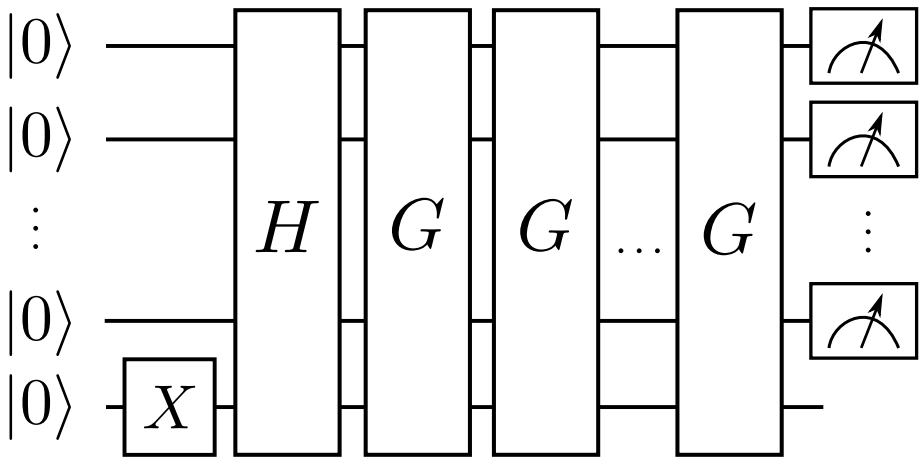
Learning outcomes

- implement Grover's algorithm in PennyLane
- define and calculate cost metrics of quantum circuits
- define and list key universal gate sets
- decompose selected single- and multi-qubit operations using one- and two-qubit universal gate sets

Implementing Grover search

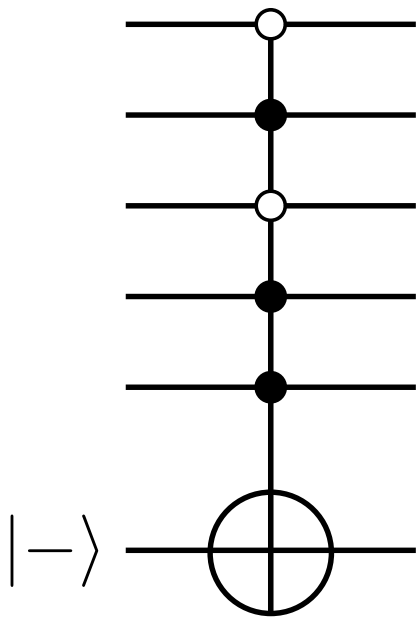
We need to design oracle and diffusion circuits. Use the "type 1" oracle:





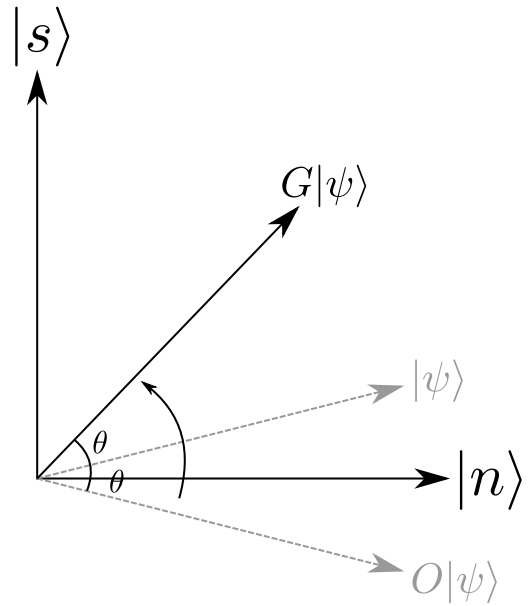
The oracle circuit

Exercise: show that a multicontrolled X gate, controlled on **s**, can be used as an oracle:



The diffusion circuit

The diffusion operator performs a reflection about the uniform superposition, $|\psi\rangle$.



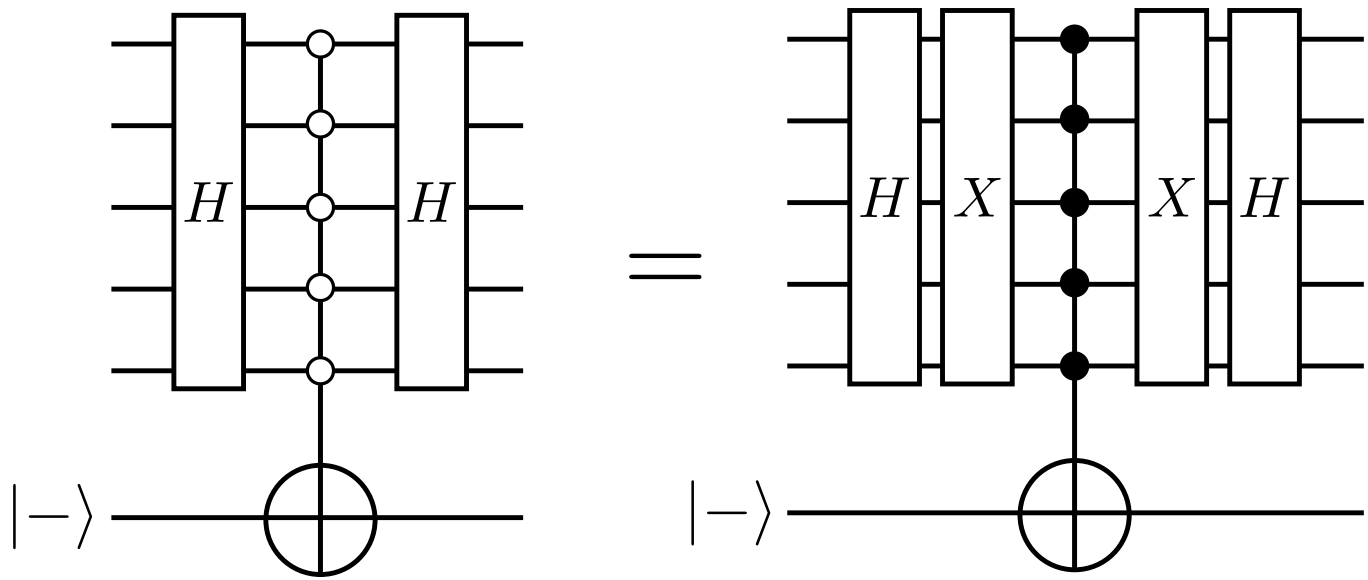
The diffusion circuit

The unitary matrix given by

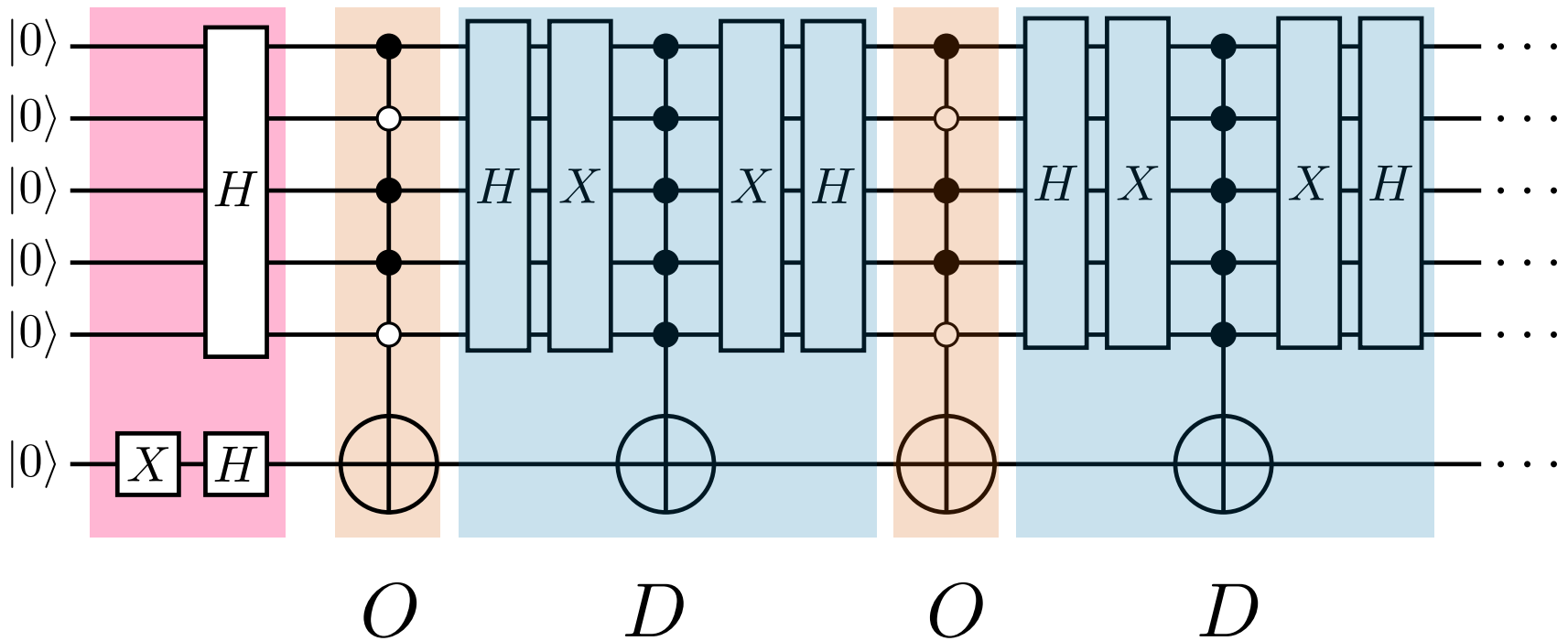
$$D = 2|\psi\rangle\langle\psi| - I$$

is equivalent to the diffusion operator.

The diffusion circuit

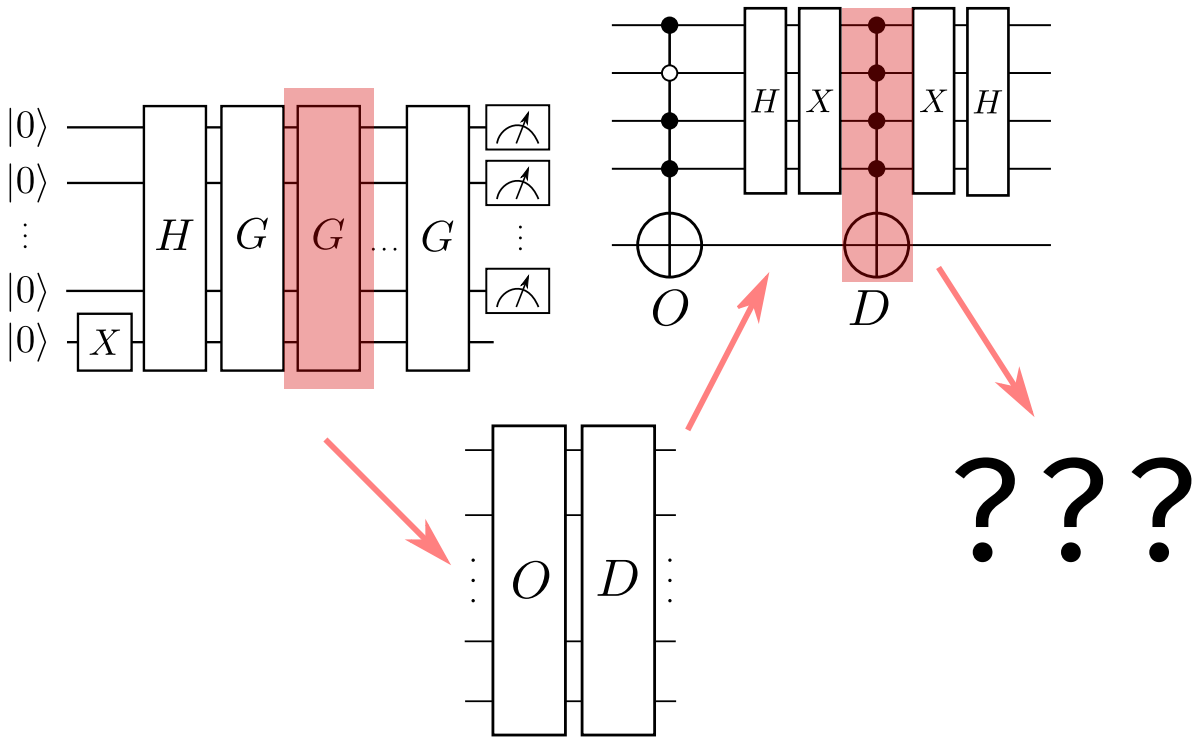


The full Grover circuit



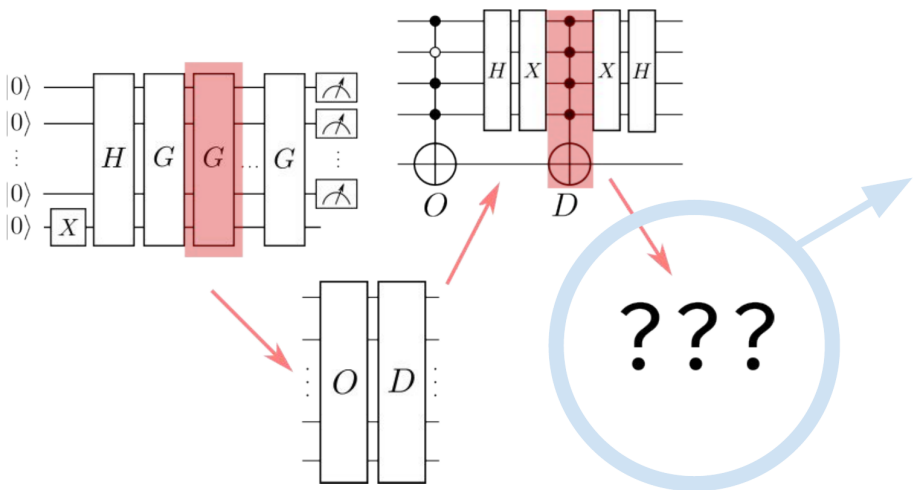
The full Grover circuit

Clearly, each of the $O(\sqrt{2^n})$ queries requires some number of gates... how much does Grover *really* cost?



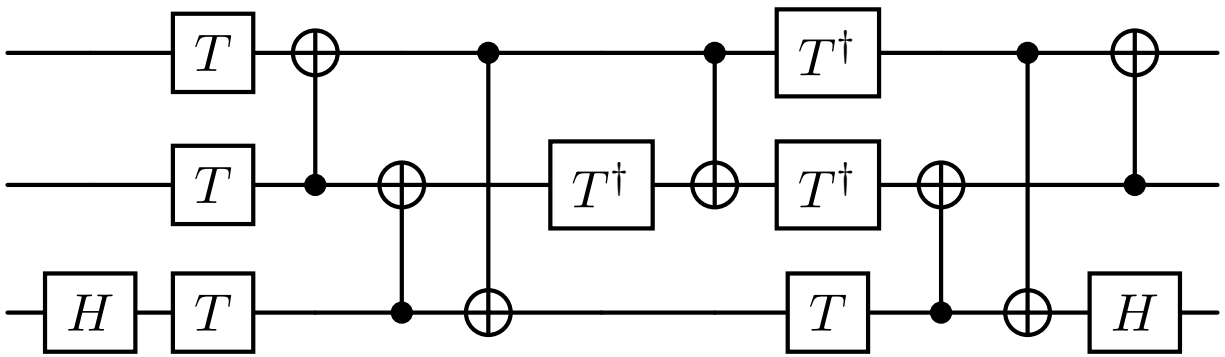
Let's open the black box!

Quantum circuit synthesis and optimization



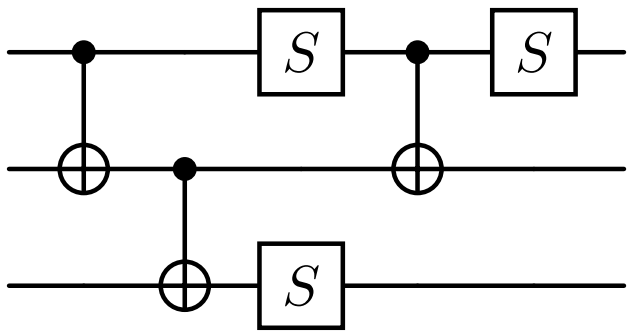
- How do we decide which gates to use?
- What criteria determine whether our choice is "good"?
- How do we improve on our choices?

Width, depth, gate count



Circuit depth

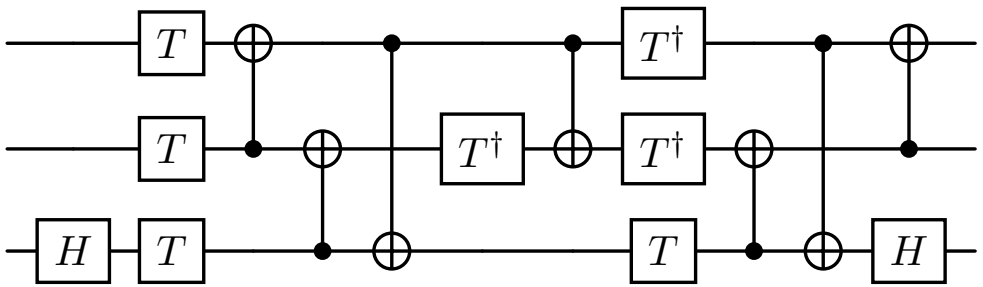
The *circuit depth* is the length of the longest path in its **directed acyclic graph (DAG)** representation.

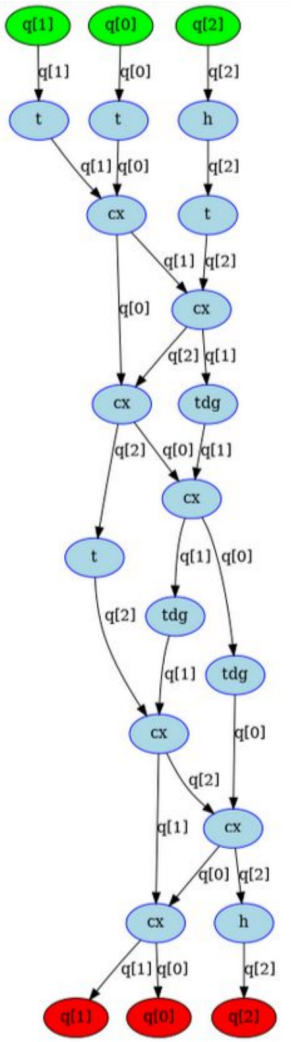


Circuit depth

Generally don't do by hand; NetworkX is your best friend.

`https://networkx.org/`

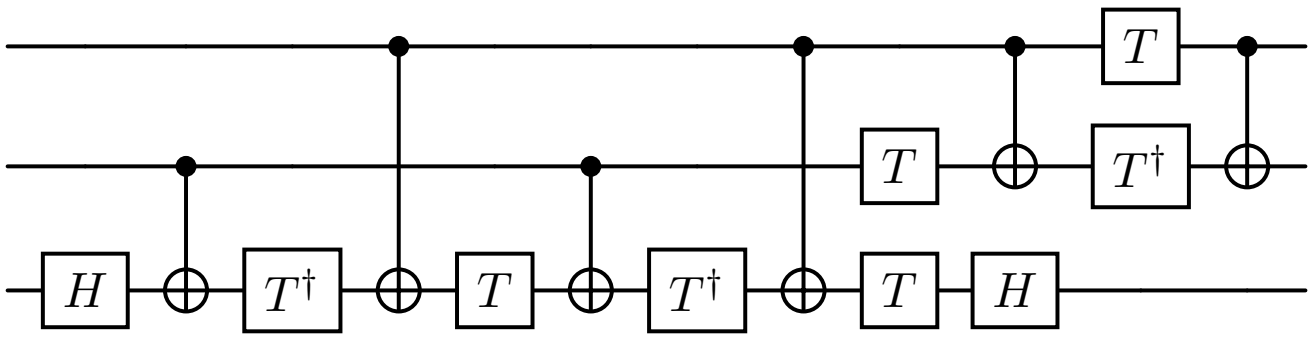




(DAG generated in Qiskit)

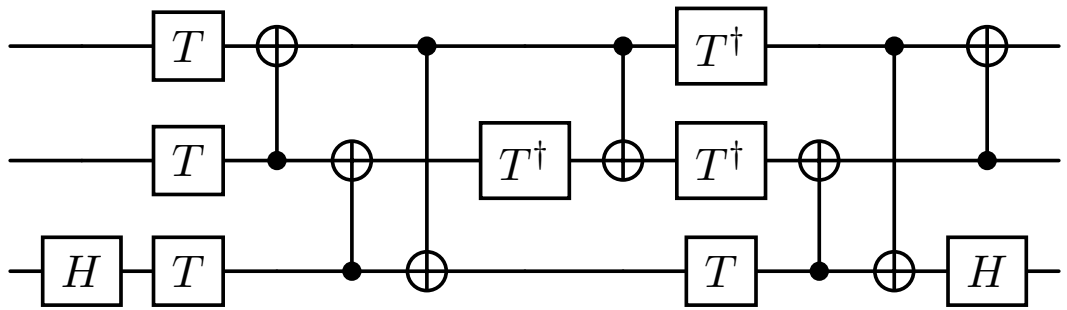
Exercise

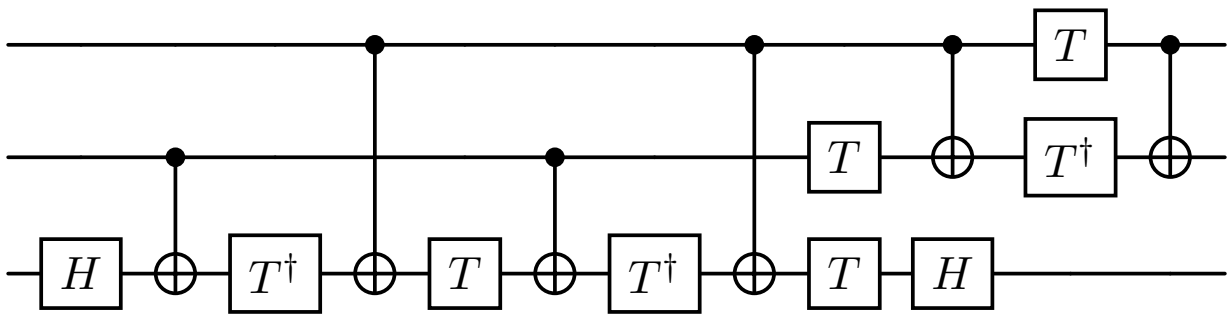
Determine resource counts of the following circuit.



Circuit synthesis and decomposition

What do these circuits do?





Circuit synthesis

Given:

- an arbitrary unitary matrix, U
- a finite set of gates, $\{G_k\}$
- a target precision, ϵ

Find

$$U' = C_1 C_2 \cdots C_m$$

such that

$$\|U - U'\| < \epsilon, \quad C_i \in \{G_k\}$$

Universal gate sets

A $\{G_k\}$ for which this can be done is called a **universal gate set**.

Examples for a single qubit:

- Any two Pauli rotations (exact)
- Hadamard and T (approximate)

Theoretical bound: $O(\log^c(1/\epsilon))$ gates, c varies from 1-4 depending on assumptions.

Universal gate sets

Using a different gate set is like *speaking a different language*.



I would like to use a real quantum computer.

I would like to use a real quantum computer.



J'aimerais utiliser un vrai ordinateur quantique.

I would like to use a real computer that is quantum.



本当の量子コンピュータを使いたい

A real quantum computer, I would like to use.

Universal gate sets

Any single-qubit unitary can be expressed using a sequence of 3 Pauli rotations around a minimum of 2 unique axes

- *ZYZ*
- *XYX*
- *XYZ*
- 9 other combinations (*Euler angles*)

qml.Rot

class Rot(*phi, theta, omega, wires, id=None*)

Bases: `pennylane.operation.Operation`

Arbitrary single qubit rotation

$$R(\phi,\theta,\omega) = RZ(\omega)RY(\theta)RZ(\phi) = \begin{bmatrix} e^{-i(\phi+\omega)/2}\cos(\theta/2) & -e^{i(\phi-\omega)/2}\sin(\theta/2) \\ e^{-i(\phi-\omega)/2}\sin(\theta/2) & e^{i(\phi+\omega)/2}\cos(\theta/2) \end{bmatrix}.$$

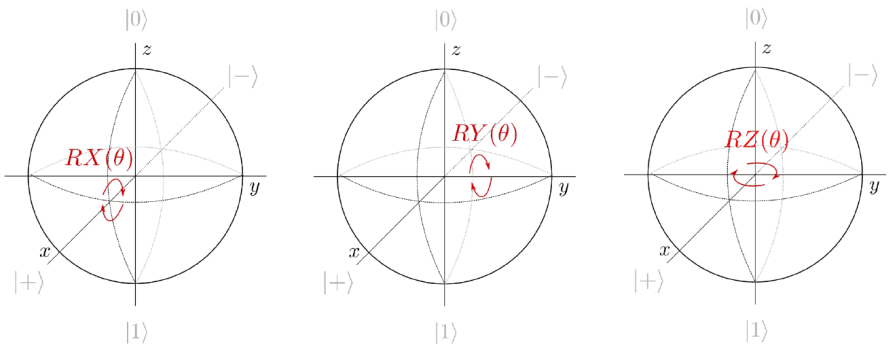
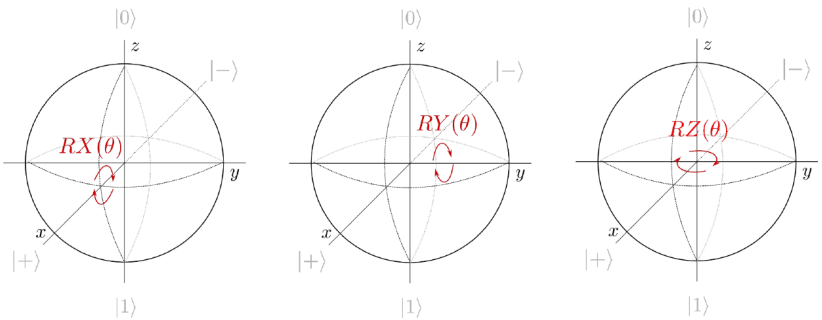


Image credit: Codebook node I.6

Hardware-specific example

RX/*RY*/*RZ* are **arbitrary-angle** rotations about **fixed axes**



IonQ's trapped ion hardware native gates: *GPI*/*GPI*2, **fixed-angle** rotations about **arbitrary axes** in *xy*-plane

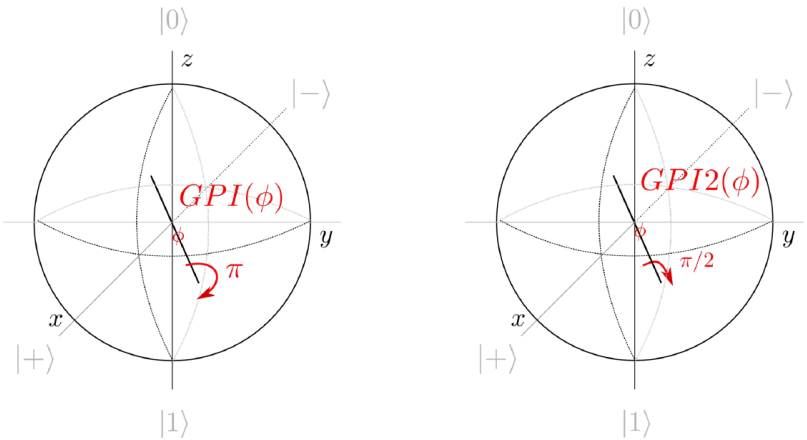
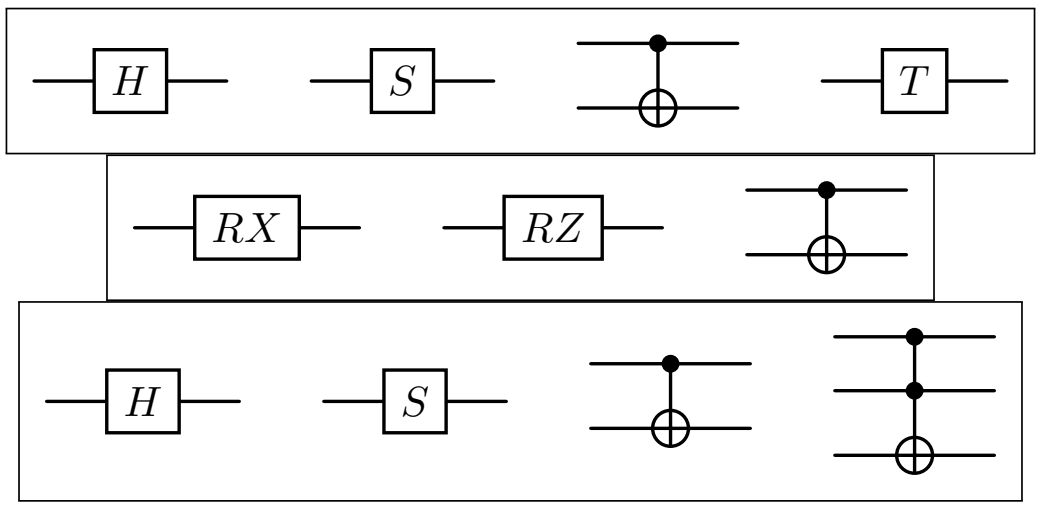


Image credit: adapted from Codebook node I.6.

See blog post: `https://pennylane.ai/blog/2023/06/the-ionizer-building-a-hardware-specific-transpiler-using-pennylane`

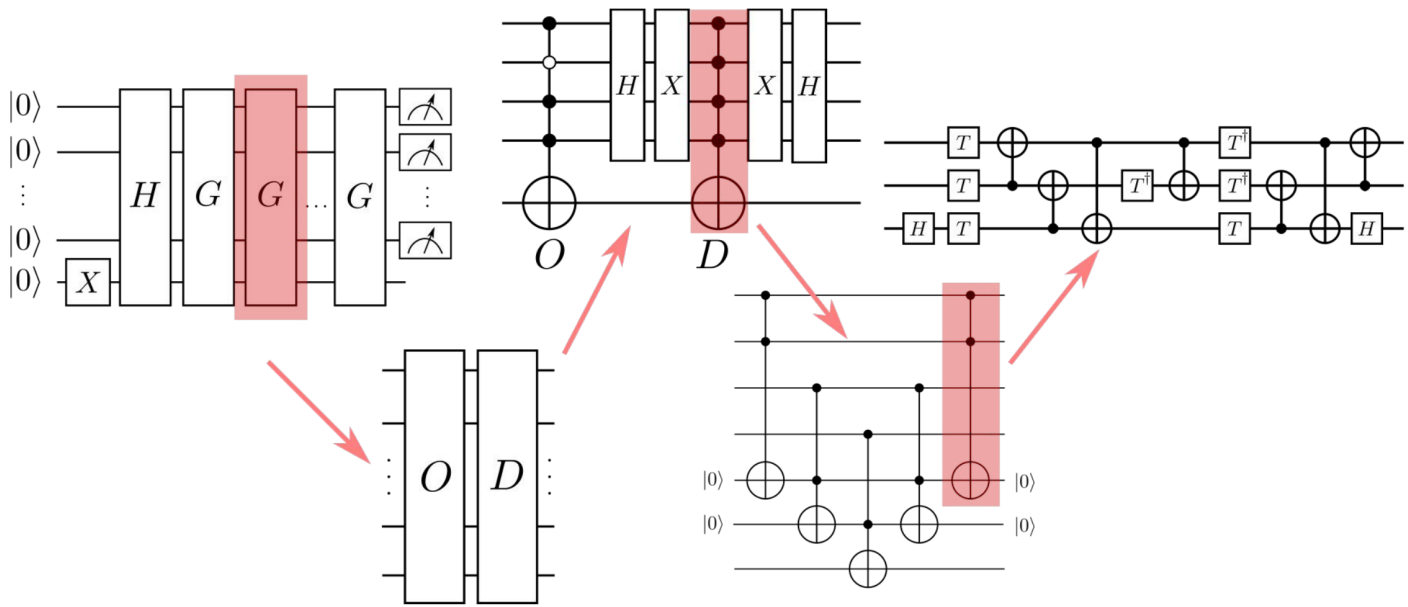
Universal gate sets

Multi-qubit gate sets



Circuit decomposition

Decompositions of many common operations are **known**. Some are even straightforward to work out by hand.



Exercise

Express Pauli X in terms of H and T.

Hint: start by expressing it using H and Z.

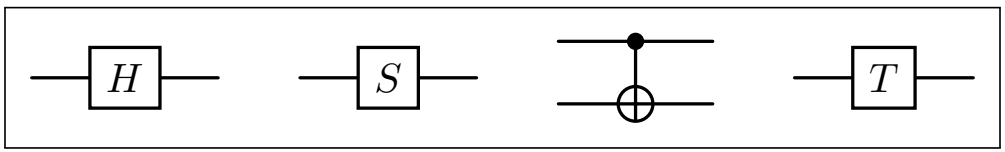
Exercise

Express Hadamard in terms of $\{RX, RY, RZ\}$

Hint: there are many ways to do this!

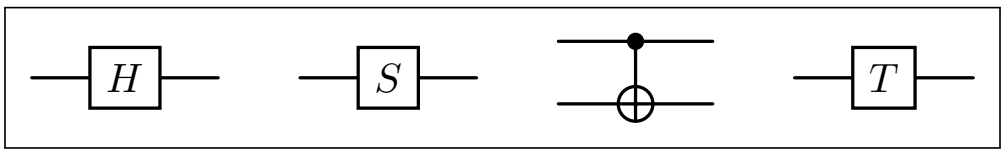
Exercise

Express CZ using gates from



Take-home exercise

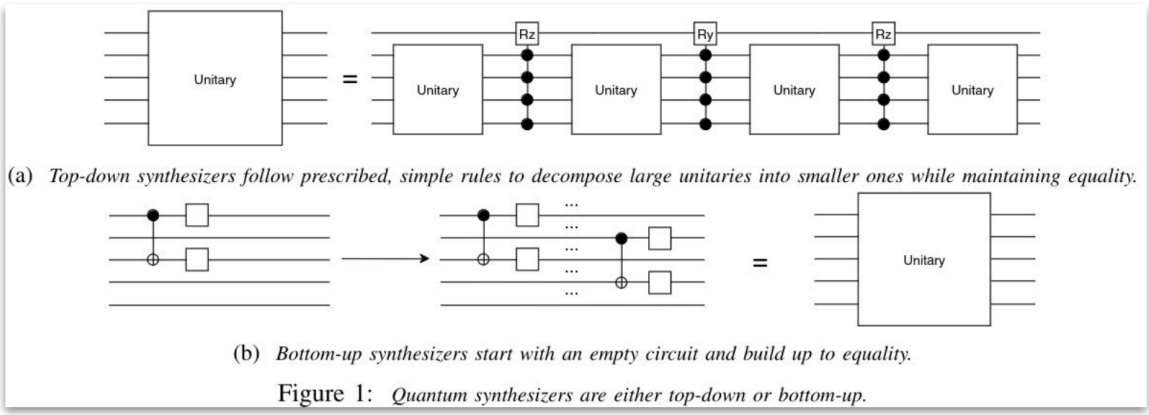
Express controlled Hadamard using gates from



Synthesis and decomposition techniques

For under-specified or arbitrary operations, or for special gate sets, we may need to **find** a decomposition.

- search problem
- number-theoretic techniques
- numerical methods



(a) *Top-down synthesizers follow prescribed, simple rules to decompose large unitaries into smaller ones while maintaining equality.*

(b) *Bottom-up synthesizers start with an empty circuit and build up to equality.*

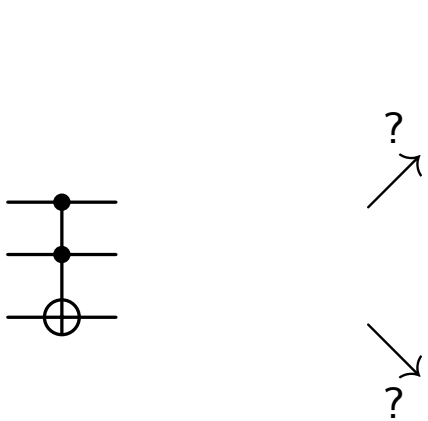
Figure 1: *Quantum synthesizers are either top-down or bottom-up.*

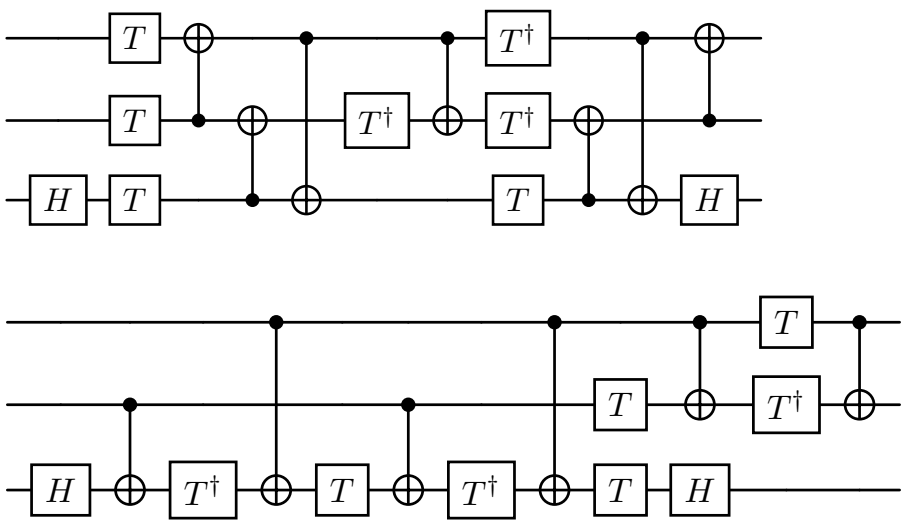
Image credit: E. Younis, K. Sen, K. Yelick, and C. Iancu. *QFAST: Conflating Search and Numerical Optimization for Scalable Quantum Circuit Synthesis*. arXiv:2103.07093 [quant-ph]

Challenges

Synthesis is hard!

- algorithm complexity and computational resources
- how to choose the best decomposition to *enable further optimization* down the line?





Challenges

Synthesis is hard!

- algorithm complexity and computational resources
- how to choose the best decomposition to *enable further optimization* down the line?
- manage tradeoffs between various resources

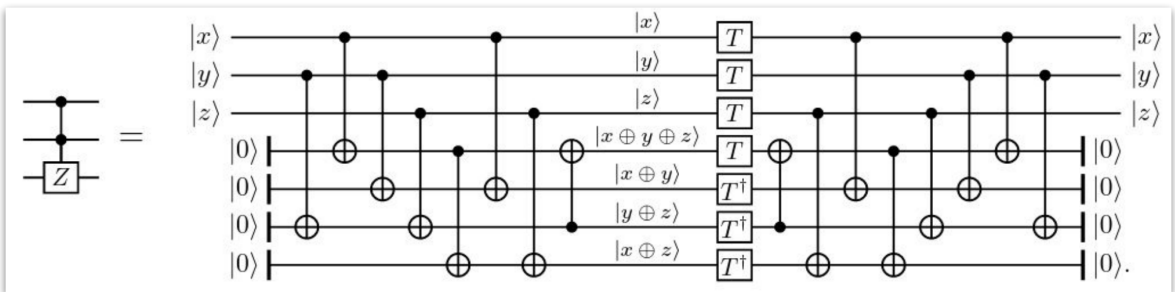


Image credit: P. Selinger (2013) *Quantum circuits of T-depth one*. Phys. Rev. A 87, 042302 (2013)

Example

Let's decompose a multi-controlled NOT.

Multi-controlled NOT decomposition strategies

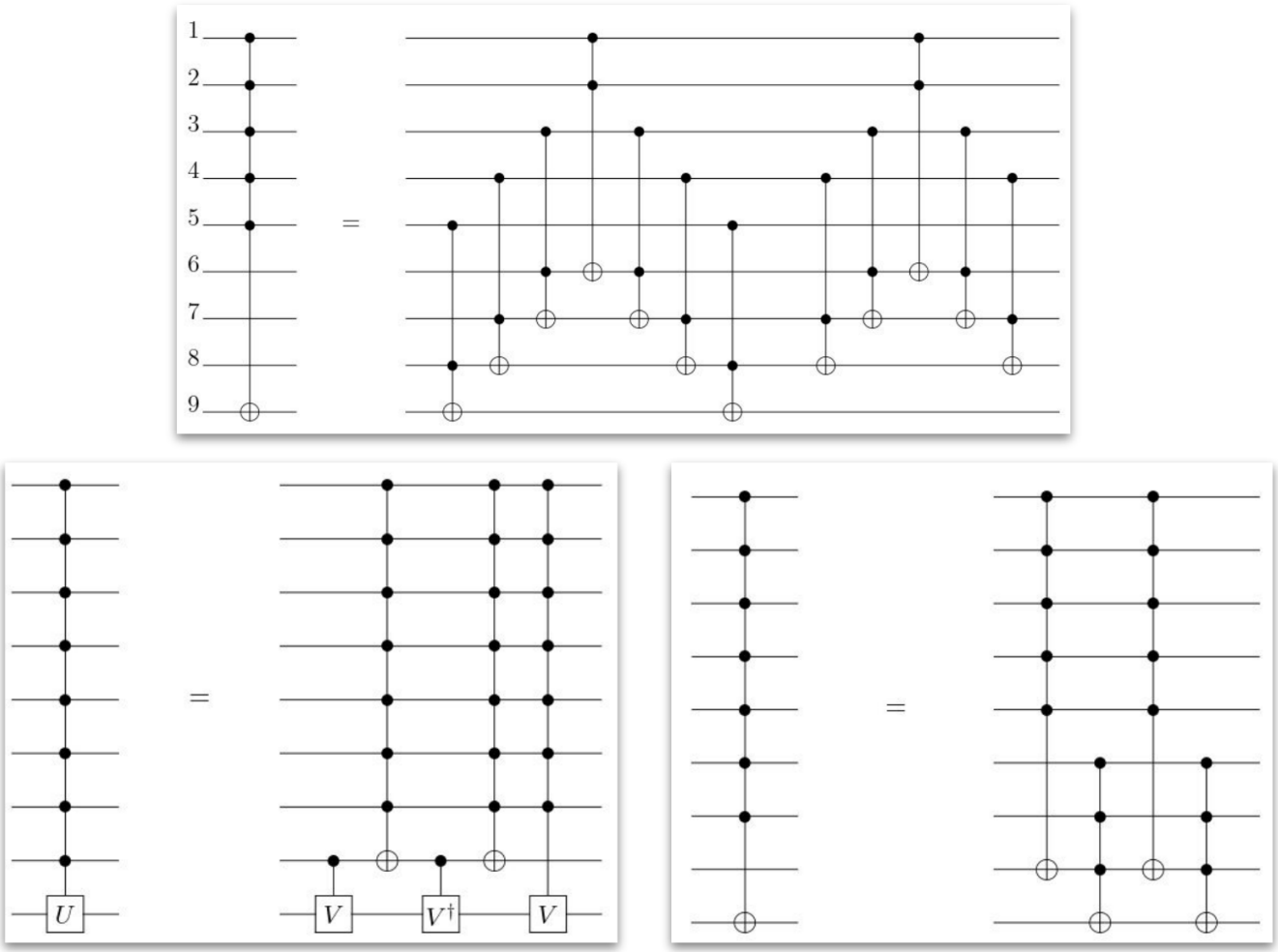


Image credit: Barenco et al., *Elementary gates for quantum computation*. Phys. Rev. A 52 (1995) 3457

Next time

Content:

- The quantum compilation stack (from top to bottom)
- Quantum circuit optimization

Action items:

1. Prepare for Quiz 5
2. Watch PrairieLearn for project information

Recommended reading:

- Nielsen & Chuang 4.5
- Compiling quantum circuits (PennyLane documentation)
- Explore the `qp.transforms` module